

History Name: GROUP3_Infection_RNASeq_Assignment3

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Galaxy History: <https://usegalaxy.org/u/celinebahian/h/group-3-infection-rnaseq-assignment3>

Complete Pipeline Steps

Step 1: Import FASTQ Reads from the SRA Database

Four RNA-seq samples were imported: Control Rep1, Control Rep2, Infected Rep1, and Infected Rep2.

Step 2: FastQC – Quality Assessment

FastQC was performed on all four samples to assess the quality of the raw sequencing reads. The outputs included quality reports and HTML summary pages.

Step 3: MultiQC – Quality Summary

MultiQC was used to combine the FastQC results from all four samples into one summary report, allowing the overall sequencing quality to be reviewed.

Step 4: Reference Genome and Annotation

The Mus musculus reference genome (Mus_musculus_v1_genome.fasta.gz) and its corresponding gene annotation file (Mus-musculus_v1_annotation.gtf.gz) were uploaded to Galaxy.

Step 5: TopHat – Genome Alignment

TopHat was used to align the RNA-seq reads to the Mus musculus reference genome. A BAM alignment file was generated for each of the four samples.

Step 6: featureCounts – Gene-level Quantification

featureCounts was used to determine the number of reads assigned to each gene. A gene-count table and summary table were generated for the samples.

Step 7: DESeq2 – Differential Gene Expression Analysis

DESeq2 was used to compare gene expression between the control and infected groups. The analysis produced a differential-expression results table, along with PCA, MA, and dispersion plots. The results table was sorted according to log2 fold change (log2FC).